SIMPLE HARMONIC MOTION - SIMPLE PENDULUM

Author(s): Tanali Bhattacharyay, Radhika Oka, Vaibhav Wagh, Sugat Dabholkar

Subject: Physics, Self-directed

Time: 80-90 minutes

Level: Class IX

Version: v2

OVERVIEW

This lesson is about understanding and investigating simple harmonic motion (SHM) of a simple pendulum. Students use a computational model of a simple pendulum to learn about its position, velocity and acceleration. Students can also observe changes in the mechanical energy (kinetic energy and potential energy) of the pendulum.

STANDARDS

NEXT GENERATION SCIENCE STANDARDS

- Physical Science
  - [HS-PS4-1] Use mathematical representations to support a claim regarding relationships among the frequency, wavelength, and speed of waves traveling in various media.

COMPUTATIONAL THINKING IN STEM

- Data Practices
  - Analyzing Data
  - Collecting Data
- Modeling and Simulation Practices
  - Assessing Computational Models
  - Using Computational Models to Find and Test Solutions
  - Using Computational Models to Understand a Concept

CREDITS

Lesson development: Tanali Bhattacharya, Radhika Oka, Vaibhav Wagh, Sugat Dabholkar

Topic selection: Tanali Bhattacharya, Radhika Oka

Model coding: Vaibhav Wagh, Sugat Dabholkar

ACKNOWLEDGEMENT

This lesson has been co-designed as a part of a co-design project by CoESME at IISER Pune and CCL at Northwestern.

ACTIVITIES

1. Introduction to computational thinking
2. Exploring a model of simple pendulum
3. Investigating motion of a pendulum
4. Designing an experiment to find 'g'
5. Understanding KE and PE of a pendulum
6. Evaluating and extending the model

STUDENT DIRECTIONS AND RESOURCES

In this lesson, you will be able to -

1) learn how to measure the amplitude and time period of a simple pendulum

2) understand the changes in displacement, velocity, and acceleration during simple harmonic motion (SHM) of a pendulum

3) understand the energy relationship between kinetic and potential energy during SHM of a pendulum

4) learn how to verify the value of acceleration due to gravity

5) investigate how the mass of the bob (m) and the length of the pendulum (l) affects the time period

6) use a computational model to perform scientific investigations

1. INTRODUCTION TO COMPUTATIONAL THINKING

The embedded resource from above can be accessed via this link (https://www.youtube.com/embed/2ACjQLXrdug)

In this video, researchers at the Indian Institute of Science Education and Research (IISER) Pune talk about how they use computational tools and methods in their work.

Scientists and Mathematicians are increasingly using computational tools and methods to understand natural phenomena and solve problems. The thinking required to do so is called computational thinking.

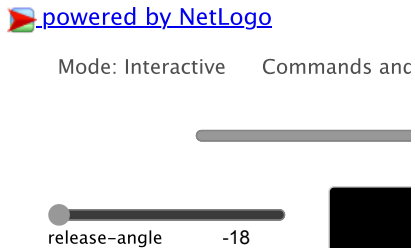
In this lesson using a computational model, we will learn computational thinking and simple harmonic motion of a pendulum.

QUESTION 1.1

Based on what you saw in the video, explain why science students need to develop computational thinking.

2. EXPLORING A MODEL OF SIMPLE PENDULUM

Explore the model first for at least 100 ticks. Then answer the questions below.



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/jamieadmin/2021/02/10/simple%20pendulum%201_2021-02-10-19-57-22-835037.html)

QUESTION 2.1

Write at least two observation/s that you find interesting in the model above.

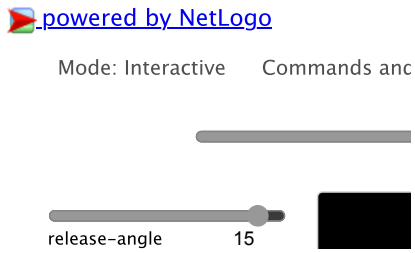
QUESTION 2.2

What parameters can you vary in the model?

QUESTION 2.3

Describe your observations regarding how the motion of pendulum changes as you vary the parameters.

3. INVESTIGATING MOTION OF A PENDULUM



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/jamieadmin/2021/02/10/simple%20pendulum%202_2021-02-10-20-00-07-289962.html)

QUESTION 3.1

Describe your observations about how the displacement from the mean position of the pendulum changes as time progresses.

QUESTION 3.2

Describe your observations about how the velocity of the pendulum changes as time progresses.

QUESTION 3.3

Describe your observations about how the acceleration of the pendulum changes as time progresses.

QUESTION 3.4

Which parameter represents time in this model?

QUESTION 3.5

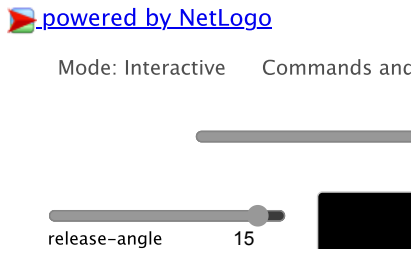
Find the time period of the pendulum.

QUESTION 3.6

How can you establish the relationship between ticks and one second?

(Hint: You will have to use the data you collected when you performed an experiment with a real pendulum. Skip this question if you have not performed the physical experiment.)

4. DESIGNING AN EXPERIMENT TO FIND 'G'



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/jamieadmin/2021/02/10/simple%20pendulum%202_2021-02-10-20-01-46-410289.html)

QUESTION 4.1

How could you conduct a computational experiment with this model to find out 'g'?

(You will need to use information from the physical experiment you already performed with a real pendulum. If you have not done the physical experiment, use your imagination to design a computational experiment that can be also performed in real life.)

QUESTION 4.2

Conduct the experiment and find the value of 'g'.

(Use the correct unit of time. You will need to convert 'ticks' to seconds.)

QUESTION 4.3

Design another computational experiment to verify that the period of the pendulum does not change if you change the release angle. Describe your experimental design.

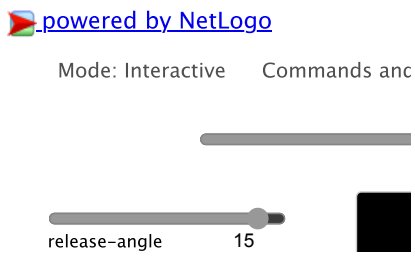
QUESTION 4.4

Perform the experiment and write your observations.

QUESTION 4.5

Write the conclusion of your experiment.

5. UNDERSTANDING KE AND PE OF A PENDULUM



The embedded resource from above can be accessed via this link (https://ct-stem.s3.amazonaws.com/uploads/jamieadmin/2021/02/10/simple%20pendulum%203_2021-02-10-20-02-49-654923.html)

QUESTION 5.1

Explore the energy (KE and PE) graph and note down your observations.

QUESTION 5.2

In the energy (KE and PE) graph, which colour represents KE and which colour represents PE?

QUESTION 5.3

Find the relationship between the position of the bob and its kinetic and potential energy.

QUESTION 5.4

Note down the position/s of the bob where its KE and PE are maximum and minimum.

6. EVALUATING AND EXTENDING THE MODEL

QUESTION 6.1

What are the advantages of using computational models for research?

QUESTION 6.2

List some limitations of the last model you used in this lesson.

QUESTION 6.3

To overcome one of those limitations, how will you modify the model?

QUESTION 6.4

What is your biggest takeaway from this lesson?

This material was developed by the

CT-STEM Project

at Northwestern University. To inquire about using our material in your school or as part of research, please contact

ctstem@northwestern.edu

.

This work was made possible through generous support from the National Science Foundation (grants CNS-1138461, CNS-1441041, DRL-1020101, DRL-1640201 and DRL-1842374) and the Spencer Foundation (Award #201600069). Any opinions, findings, or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the funding organizations.

© 2026 CT-STEM Project. Some rights reserved.